347631

Course Name: MACHINE LEARNING

Course Outcome:

CO1: Understand the concepts of Machine learning and intelligent agents.
CO2: Understand and learn knowledge representation and reasoning for the problem-solving solving.
CO3: Apply basic search techniques for problem-solving.
CO4: Understand and apply learning techniques.
CO5: Apply and utilize AI knowledge for application in the real world

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Mid Term Examination, Odd Semester 2024-25

B. Tech. CS, 3rd Year, V Semester

BCSE0105: MACHINE LEARNING

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	You are working with a healthcare dataset that contains patient information, but several rows have missing values for the Age and Blood Pressure columns. What are some strategies for handling missing values in these columns, and how would you decide whether to use mean imputation, median imputation, or deletion of missing data points?	3	2	A	C
2	Your dataset has a Country feature with the values USA, UK, and India. You want to use this feature in a machine learning model. What are the different encoding techniques you can apply to this categorical feature? Explain the differences between one-hot encoding and label encoding and when each method is appropriate.	3	2	A	C
3	You are working on a dataset that contains information about employees, including features like their age, experience, and annual income. The goal is to predict whether an employee will stay with the company for more than 2 years (Yes/No). Is this a regression or classification problem? Justify your answer and explain how the model's output would differ in each case.	3	2	A	P
4	You are working on a dataset containing information about customer transactions at a retail store. You need to predict both how much each customer will spend (a continuous variable) and whether they will buy a specific product (binary outcome).	3	1	R	F

	How would you handle feature engineering for these two tasks? Would you preprocess the data differently for a regression model than for a classification model? Give examples of how feature transformations might differ.				
5	You are using a dataset of house prices, but you observe a few houses that are priced much higher than the rest of the data (outliers). What techniques would you use to handle these outliers, and how would their presence affect your model's performance if left untreated? When should you remove outliers versus applying transformation methods?	3	3	A	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	You are given a dataset of 10 examples with 3 binary features each (1 for positive, 0 for negative). The task is to learn a concept that classifies the examples as "True" or "False". The target concept is a conjunction of the three features. DATA: Given the following data: Example 1: (1, 0, 1) → True Example 2: (0, 1, 1) → False Example 3: (1, 1, 0) → True Find the most specific hypothesis that fits the data.	5	4	U	P
7	You are using a linear regression model to predict house prices based on the following data: Size of house (in square feet): [1000, 1500, 2000, 2500] Price of house (in $1000): [300, 400, 500, 600] Fit a linear regression model to this data and find the equation of the line (i.e., determine the slope and intercept).	5	4	U	P
8	You have a dataset of 1000 records with 10 features. One feature, Age, is missing in 5% of the cases. The mean Age for the available data is 30 years, and the standard deviation is 5 years.If you decide to impute the missing Age values using mean imputation, what will be the new mean of the Age feature?	5	3	R	F